UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination

Engineering Mathematics-II (BAS203)

Roll No. _______________

Subject: Engineering Mathematics-II **Time:** 3 Hours **Max. Marks:** 70

Code: BAS203

General Instructions:

1. Attempt all questions.
2. Show all necessary steps in calculations.
3. Use of scientific calculator is permitted.

SECTION A — SECTION A

Short Answer Questions

2 marks each | Attempt all questions

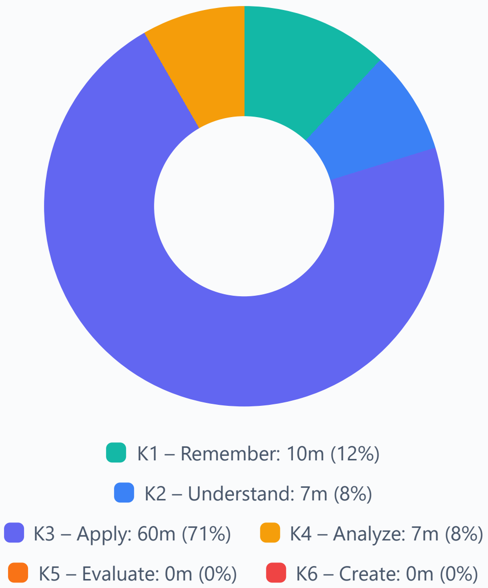
Q.No.	Question	Marks	CO
1	Define the substitution required to transform a Cauchy-Euler equation into a linear differential equation with constant coefficients.	**2**	**CO1**
2	State the necessary conditions for the existence of the Laplace transform of a function f(t).	**2**	**CO2**
3	State the D’Alembert’s ratio test for the convergence of a series of positive terms.	**2**	**CO3**
4	Write the Cauchy-Riemann equations for a function $f(z) = u + iv$ in polar coordinates (r, θ).	**2**	**CO4**
5	Define a simple pole and write the formula for calculating the residue of a function at a simple pole.	**2**	**CO5**
6	Solve the second-order linear differential equation: $y'' - 5y' + 6y = 0$.	**2**	**CO1**
7	Find the Laplace transform of f(t) = e^{2t} sin(3t) using the shifting property.	**2**	**CO2**

SECTION B — SECTION B			
Long Answer Questions			
7 marks each | Attempt all questions			
Q.No.	**Question**	**Marks**	**CO**
8	Test the convergence of the series: $\Sigma [(n!)^2 / (2n)!] x^n$ for positive values of x.	**7**	**CO3**
9	If $u = x^3 - 3xy^2$, verify that u is harmonic and find the corresponding analytic function $f(z) = u + iv$ using the Milne-Thompson method.	**7**	**CO4**
10	Evaluate the integral $\oint [(z^2 + 1) / (z - 1)^2 (z + 2)] dz$ around the circle $\|z\| = 3$ using the Cauchy Residue Theorem.	**7**	**CO5**
11	Solve the differential equation $y'' + 4y = \tan(2x)$ using the Method of Variation of Parameters.	**7**	**CO1**
12	Solve the initial value problem $y'' + 2y' + 5y = e^{-t} \sin(t)$, given $y(0) = 0$ and $y'(0) = 1$, using the Laplace transform method.	**7**	**CO2**

SECTION C — SECTION C			
Detailed Answer Questions			
7 marks each | Attempt all questions			
Q.No.	**Question**	**Marks**	**CO**
13	**(a)** Obtain the Fourier series for the function $f(x) = x^2$ in the interval $(-\pi, \pi)$.	**7**	**CO3**
	OR		
	(b) Find the half-range cosine series for the function $f(x) = \sin(x)$ in the interval $(0, \pi)$.	**7**	**CO3**
14	**(a)** Explain the properties of Mobius transformations and find the transformation that maps the points $z = 0, 1, \infty$ onto $w = i, 1, -i$.	**7**	**CO4**
	OR		
	(b) Discuss the analyticity of the function $f(z) = \|z\|^2$ and determine the points where the derivative exists.	**7**	**CO4**
15	**(a)** Expand the function $f(z) = 1 / [(z + 1)(z + 3)]$ in a Laurent series valid for the region $1 < \|z\| < 3$.	**7**	**CO5**
	OR		
	(b) Evaluate the integral $\oint [\sin(\pi z^2) + \cos(\pi z^2)] / [(z - 1)(z - 2)]\, dz$ around the circle $\|z\| = 3$ using Cauchy's Integral Formula.	**7**	**CO5**
16	**(a)** Solve the simultaneous linear differential equations: $dx/dt + 2x + 3y = 0$ and $dy/dt + 3x + 2y = 0$, given $x(0) = 1$ and $y(0) = 0$.	**7**	**CO1**
	OR		
	(b) Solve the differential equation $x^2 y'' - xy' + y = \ln(x)$ using the Cauchy-Euler method.	**7**	**CO1**
17	**(a)** State and prove the Convolution Theorem for Laplace transforms and use it to find $L^{-1} \{ 1 / [s^2 (s^2 + 1)] \}$.	**7**	**CO2**
	OR		
	(b) Find the Laplace transform of a periodic square wave function $f(t)$ with period $2a$, defined as $f(t) = E$ for $0 < t < a$ and $f(t) = -E$ for $a < t < 2a$.	**7**	**CO2**

Question Paper Analysis

Bloom's Taxonomy Distribution



Course Outcome Distribution